Work sheet - 06

Course: Aptitude Topic: Percentage

1. 20% of 300
   a) 60 b) 70 c) 80 d) 90
   Sol:20% of 300=20/100×300=60

2. Express 25% in fraction.
   a) 1/3 **b) 1/4** c) 1/5 d) 1/6
   Sol: 25%=25/100=1/4

3. Convert 3/8 into percentage.
   a) 12.5% b) 25% **c) 37.5%** d) 50%
   Sol:3/8×100=37.5%

4. 60 kg is what percent of 240 kg?
   a) 25% b) 50% c) 75% d) 12.5%
   Sol: 60/240×100=25%

5. What fraction is equivalent to 15%?
   a) 3/20 b) 7/20 c) 1/20 d) 2/15
   Sol: 15%=15/100=3/20

6. 60% of a number is 84. What is the number?
   a) 100 **b) 140** c) 70 d) 50
   Sol: 60% of x=84
   60/100× x=84
   X=140

7. 30% of 140 = x% of 840. Find x.
   a) 5 b) 10 c) 15 d) 20
   Sol: 30/100×140=X/100×840
   X=5

8. A number decreased by 15% gives 51. Find the number.
   a) 46 **b) 60** c) 30 d) 59
   Sol:
   After 15% decrease, number becomes 85%.
   85%=51
   1%=51/85

   100%=51/85x100 =60

9. 40% of a number is more than 20% of 650 by 190. Find the number.
   a) 600 b) 700 **c) 800** d) 900
   Sol:
   40% of A-20% of 650=190
   40/100x A = 190+130
   A= 32x100/40
   A= 800

10. First two numbers are 25% and 75% less than the third number. How much percent is the second number less than the first?
a) 65⅔% **b) 66⅔%** c) 66⅓% d) 67⅔%
Sol:
Let the third number = 100
First number = 75
Second number = 25
Difference = 50
50/75×100=200/3=66 2/3%

11. 60% of a number is added to 120, the result is the same number. Find the number.
a) 200 **b) 300** c) 400 d) 500
Sol:
60% of A +120 = A
60/100 x A +120 = A
3A/5=A-120
3A=5A-600
2A=600
A=300

12. The population of a village decreased from 4000 to 3500. Find the decreased population percentage.
a) 12½% b) 25% c) 14⅔% d) 33⅓%
Sol:
Decrease = 500
500/4000×100=12.5% = 12½% decrease

13. The price of an article increased from 15000 to 20000. Find the increased percentage.
a) 50% **b) 33⅓%** c) 25% d) 20%
Sol:
Increase = 5000
5000/15000×100=33⅓%

14. A student has to get 35% of the maximum marks to pass. He got 172 marks and failed by 3 marks. Find the maximum marks.
a) 600 b) 700 c) 400 **d) 500**
Sol:
Let the maximum marks be X
Now 35% of X = 172 + 3
35/100 x X =175
X = 500
Alternative Method
Pass marks = 175 marks
Pass % = 35%
35% = 175
100% = 175/35 x 100 = 500

15. In an election, one of the two candidates gets 42% of the total votes and still loses by 368 votes. Find the total number of votes.
a) 3200 **b) 2300** c) 1300 d) 1600

Sol:
Difference = 58% − 42% = 16%
16%=368
100% = 368/16 x 100 = 2300

16. In a fraction, the numerator is increased by 15% and the denominator decreased by 25%. The value of the fraction becomes 3/8. Find the original fraction.
a) 45/184 b) 184/45 c) 35/124 d) 124/35
Sol:
Let the original fraction be x/y 115% of x / 75% of y=3/8
115x / 75y=3/8
x/y=45/184
Answer: 45/184

17. If A's income is 20% more than B's income, what percent is B's income less than A's income?
a) 33⅓% b) 25% c) 20% **d) 16⅔%**
Sol:
Let B's income = 100
A's income = 120
Difference = 20
20/120×100= 16 2/3%

18. If A's salary is 25% less than B's salary, then what percent is B's salary more than A's salary?
a) 33⅓% b) 25% c) 20% d) 16⅔%
Sol:
Let the salary of B be 100.
Then the salary of A = 75
Difference = 25
25/75×100= 33⅓%

19. The population of a town is 80,000. If it increases annually at the rate of 10% per annum, what will be its population after 3 years?
a) 2,06,480 b) 2,06,680 c) 1,06,680 **d) 1,06,480**
Sol:
Population after 3 years:
=80000(1+10/100)^3
=80000(11/10)^3
=80000 x 1331/1000
=1,06,480

Alternative Method:
1st year: 80,000 + 10% = 88,000
2nd year: 88,000 + 10% = 96,800
3rd year: 96,800 + 10% = 1,06,480
Answer: 1,06,480

20. If the price of sugar is increased by 50%, find how much percent a householder must reduce the consumption of sugar so as not to increase the expenditure.
a) 20% b) 25% **c) 33⅓%** d) 50%
Sol:
Original price: ₹100 for 3 kg
Increased price: ₹150 for 3 kg
For ₹100, sugar obtained = 2 kg
Reduction = 1 kg out of 3 kg
1/3×100=33⅓%
Answer: 33⅓%

21. If the price of tea falls by 20%, by how much percent must a householder increase consumption so as not to decrease the expenditure?
a) 20% **b) 25%** c) 33⅓% d) 50%
Sol:
Original price: ₹100 for 4 kg
Reduced price: ₹80 for 4 kg
For ₹100, quantity obtained = 5 kg
Increase = 1 kg out of 4 kg
1/4×100=25%
Answer: 25%

22. On the reduction of the price of apples, a man can purchase 24 apples more in the same amount in which he could purchase 126 apples at the previous rate. What is the percentage reduction in the price?
a) 16% b) 20% c) 24% d) 30%
Sol:
Let the price of 126 apples = Rs. x
Price of 1 apple = x / 126
Percentage reduction
=((x/126−x/150) / x/126)×100%
By solving, we get:
Percentage reduction = 16%

Alternative Method:
Let the price of one apple be Rs.1
Then: Rs.126 = 126 apples
Rs.150 = 150 apples
After decrease: Rs.126 = 150 apples
Increase in apples = 24
24 / 150×100=16% decrease

23. One litre of water is evaporated from 6 litres of a solution containing 5% salt. Find the percentage of salt in the remaining solution.
a) 5% **b) 6%** c) 7% d) 8%
Sol:
Salt in 6 litres = 5% of 6 = 0.3 litres
Water = 5.7 litres
After evaporation: Water = 4.7 litres , Salt = 0.3 litres

Total solution = 5 litres
0.3/5×100=6%
Answer: 6%

24. A mixture of 40 litres of milk and water contains 10% water. How much water should be added so that water becomes 20%?
a) 5 litres b) 6 litres c) 7 litres d) 8 litres
Sol:
10% of 40 = 4 litres (water)
Let x litres of water be added.
(4+x / 40+x)×100=20
Solving, x = 5 litres
Answer: 5 litres

25. The monthly income of a person is Rs. 8000. If his income is increased by 20%, what will be his new monthly income?
a) 7000 b) 6400 c) 8200 **d) 9600**
Sol:
20% of 8000 = 1600
New income = 8000 + 1600 = Rs. 9600

26. The salary of a worker is first increased by 5% and then decreased by 5%. What is the net percentage change in salary?
a) 0.60% b) 0.50% **c) 0.25%** d) 0.75%
Sol:
According to formula: Decrease % = a^2 / 100
=5^2/100 % =0.25%
Net change = 0.25% decrease

27. The price of an article is first increased by 20% and later decreased by 25%. Find the percentage change in the final price.
a) 10% decrease b) 10% increase c) 20% decrease d) 20% increase
Sol:
Let original price = 100
After 20% increase → 120
After 25% decrease → 90
Decrease = 10
10/100×100=10%
Answer: 10% decrease

28. A student has to score 30% marks to pass the examination. If he got 30 marks and failed by 30 marks, find the maximum marks.
a) 100 **b) 200** c) 300 d) 400
Sol:
Pass % = 30%
Pass marks = 30 + 30 = 60
30% = 60
100%=60/30×100=200
Maximum marks = 200

29. A candidate scores 25% and fails by 60 marks. Another candidate scores 50% and gets 40 marks more than the pass marks. Find the maximum marks of the examination.
a) 400 b) 500 c) 700 d) 600
Sol:
Let maximum marks = x
25x/100+60=50x/100-40
Solving, x = 400

30. Due to 10% decrease in the price of sugar, a man can buy 5 kg more sugar for Rs.100. Find the actual price of sugar per kg.
a) 1 2/9 **b) 2 2/9** c) 3 2/9 d) 4 2/9
Sol:
Let actual price = Rs. y per kg
Before decrease: Sugar bought = 100 / y
After 10% decrease: New price = 9y / 10
Sugar bought = 1000 / 9y
1000/9y−100/y=5
1000-900 / 9y =5
Solving, y = 20/9 = 2 2/9 rupees

31. A man loses 12½% of his money and after spending 70% of the remainder, he has Rs.210 left. Find the amount he had initially.
a) 800 b) 900 c) 700 d) 600
Sol:
Remaining after loss = 87½%
Remaining after spending = 30%
30/100×175/200×x=210
x = 800

32. The price of an article is reduced by 26%. To restore the original price, by what percentage must the new price be increased?
a) 34.135% b) 32.135% c) 31.135% **d) 35.135%**
Sol:
Original price = 100
Reduced price = 74
26/74×100=35.135%
Answer: 35.135%

33. A single discount is equivalent to a discount series of 10%, 20%, and 30%.
a) 50.4% **b) 49.6%** c) 39.6% d) 60.4%
Sol:
Single discount
=100−(90%×80%×70%)
=100−50.4
=49.6%
Answer: 49.6%

34. In an examination, 35% of the total students failed in Hindi, 45% failed in English, and 20% failed in both. Find the percentage of those who passed in both the subjects.
a) 20% b) 30% **c) 40%** d) 50%
Sol:
Let A and B be the sets of students who failed in Hindi and English respectively.
$n(A) = 35$ $n(B) = 45$ $n(A \cap B) = 20$
Using the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 35 + 45 - 20 = 60$
So, percentage of students who failed in Hindi or English or both = 60%
Hence, percentage of students who passed in both subjects
$= 100\% - 60\% = 40\%$
Answer: 40%

Alternative Method:
Let the total number of students = 100

Subject	Passed	Failed
Hindi	65	35
English	55	45
Hindi & English	?	20

Total failed students = 20 (failed in both) + 15 (failed only in Hindi) + 25 (failed only in English) = 60
Total passed students = 100 − 60 = 40
Pass percentage=40/100×100=40% Pass percentage

PRACTICE QUESTIONS

1. The ratio 5:4 expressed as a percent equals:

a) 12.5% b) 40% c) 80% d) 125%

2. 45% of 300 + √? = 56% of 750 − 10% of 250

a) 60 b) 130 c) 260 d) 7600

3. One-eighth of a number is 41.5. What will 69% of that number be?

a) 219.12 b) 225.76 c) 229.08 d) 232.4

4. What percent of 88 is 33?

a) 34.5% b) 35.5% c) 36.5% d) 37.5%

5. 0.01 is what percent of 0.1?

a) 1/100 b) 1/10 c) 10 d) 100

6. 30% of apples out of 450 are rotten. How many apples are in good condition?

a) 125 b) 180 c) 240 d) 315

7. A shopkeeper sells notebooks at Rs.45 each and earns a commission of 4%. He also sells pencil boxes at Rs.80 each and earns a commission of 20%. How much commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day?

   a) 1496 b) 1586 c) 1596 d) None of these

8. In a class of 65 students and 4 teachers, each student got sweets that are 20% of the total number of students and each teacher got sweets that are 40% of the total number of students. How many sweets are there?

   a) 704 b) 845 c) 949 d) 897

9. 15% of 45% of a number is 105.3. What is 24% of that number?

   a) 374.4 b) 375 c) 385.5 d) 390

10. In an examination X secures 58% marks and Y secures 105 marks less than X. If the maximum marks were 700, what percent of marks did Y secure?

    a) 42% b) 43% c) 52% d) None of these

11. The difference between 38% of a number and 24% of the same number is Rs.135.10. What is 40% of that number?

    a) 370 b) 378 c) 386 d) 394

12. If 75% of a number is added to 75, then the result is the number itself. Find the number.

    a) 50 b) 60 c) 300 d) 400

13. A number reduced by 25% becomes 225. What percent should it be increased so that it becomes 390?

    a) 25% b) 30% c) 35% d) 45%

14. A person who spends 66 2/3% of his income is able to save Rs.1200 per month. His monthly expenses are:

    a) 1200 b) 2400 c) 3000 d) 3200

15. Ravi's expense is 30% more than Vijay's and Vijay's expense is 10% less than Sekhar's. If the sum of their expenses is Rs.6447, what would be Ravi's expense?

    a) 1890 b) 2100 c) 2200 d) 2457

16. If a number is reduced by 40%, it becomes two-thirds of another number. What is the ratio of the first number to the second number?

    a) 8:9 b) 9:8 c) 10:9 d) 9:10

17. In an election, a total of 5,00,000 voters participated. A candidate got 2,55,000 votes which was 60% of the total valid votes. What was the percentage of invalid votes?

a) 10% b) 12% c) 15% d) 300/17%

18. A student secures 90%, 60% and 54% marks in test papers with 100, 150 and 200 respectively as maximum marks. The percentage of his aggregate is:

a) 64 b) 68 c) 70 d) None of these

19. The contents of a certain box consist of 14 apples and 23 oranges. How many oranges must be removed so that 70% of the pieces of fruit in the box will be apples?

a) 6 b) 12 c) 17 d) 36

20. 5 kg of tea and 8 kg of sugar together cost Rs.172. The price of tea has risen by 20% and that of sugar by 10%. Hence the same quantities now cost Rs.199.20. What is the original price of tea per kg?

a) Rs.16 b) Rs.18 c) Rs.19 d) Rs.20

21. Sandeep spends 30% of his monthly income on food, 40% of the remaining on conveyance and clothes and saves 50% of the remaining. If his monthly salary is Rs.18,400, how much money does he save every month?

a) Rs.3624 b) Rs.3884 c) Rs.4284 d) Rs.5888

22. On a test consisting of 250 questions, Anu answered 40% of the first 125 questions correctly. What percent of the other 125 questions does she need to answer correctly for her grade on the entire exam to be 60%?

a) 60 b) 75 c) 80 d) Cannot be determined

23. A scored 30% marks and failed by 15 marks. B scored 40% marks and obtained 35 marks more than those required to pass. The pass percentage is:

a) 33% b) 38% c) 43% d) 48%

24. If the price of a book is first decreased by 25% and then increased by 20%, then the net change in the price will be:

a) No change b) 5% increase c) 5% decrease d) 10% decrease

25. If the price of a commodity is decreased by 20% and its consumption is increased by 20%, what will be the increase or decrease in the expenditure on the commodity?

a) 4% increase b) 4% decrease c) 8% increase d) 8% decrease

ANSWER KEY

1-D

2-D

3-C

4-D

5-C

6-D

7-C

8-C

9-A

10-B

11-C

12-C

13-B

14-B

15-D

16-C

17-C

18-A

19-C

20-D

21-B

22-C

23-A

24-D

25-B